Constraint-Based Control of Boiler Efficiency: A Data-Mining Approach

The original paper contains 19 sections, with 10 passages identified by our machine learning algorithms as central to this paper.

Paper Summary

SUMMARY PASSAGE 1

A. Application Background

For example, the megawatt load is determined by the demand for electricity. When increasing the boiler combustion efficiency, the megawatt load has to be respected. Unit heat rate is an important performance index for the boiler-turbine system.

SUMMARY PASSAGE 2

B. Clustering-Based Control Model

Given a system status at time described by the triplet , a centroid is denoted as when the controllable and noncontrollable parameters are used for clustering and constructing the centroids, where is the centroid part for the controllable variables, and is the centroid part for noncontrollable variables. Definitions of how to calculate the centroids are shown later. Whether changing the controllable variables from to would lead the outputs to is determined by the difference between the noncontrollable variables and .

SUMMARY PASSAGE 3

Definition 3.3:

The clustering algorithm for forming centroids and controlling the test data set is formulated next (Algorithm 1). In Step 3, the nearest centroid is determined to produce controllable variables of optimizing boiler efficiency. If the boiler efficiency corresponding to is greater than the specified efficiency level , there is no need to adjust controllable variables.

SUMMARY PASSAGE 4

End

The controlled data set is evaluated whether the target output has reached the desired value or at least has increased to a certain level. Previous research [16] has shown that boiler efficiency can be optimized with the clustering-derived centroids. The focus of this paper is on how to constrain some response variables.

SUMMARY PASSAGE 5

V. Decoupling By Selection Of Variables For Clustering

Clustering such data points is referred to as Type 1 clustering, and data vector is denoted as ; in later discussion, superscript "1" indicates Type 1, and superscript "2" indicates Type 2. Centroids generated from Type 1 clusters may lead to coupling effects with the remaining response variables because the response variables are ignored. Note that of (5.1) includes two terms, one related to controllable variables and the other related to noncontrollable ones.

SUMMARY PASSAGE 6

Vi. Industrial Case Study

The goal of the study was to maximize combustion efficiency by adjusting 18 controllable variables without impact on two response variables (i.e., megawatt load and unit heat rate selected by the domain experts). In other words, the increase in boiler efficiency should not change the megawatt load or increase the unit heat rate.

SUMMARY PASSAGE 7

C. Comparison Of The Decision-Tree Algorithm Accuracy For Different Subset Of Variables

As the number of input variables (controllable and noncontrollable) has been reduced to , it is interesting to see whether the 37 input variables provide sufficient information to predict the target output (i.e., boiler efficiency). A decision-tree algorithm [21] was applied to predict the categorized boiler efficiency by using the 37 input variables as predictors. The prediction accuracy and tenfold cross validation accuracy results are compared with the ones generated by the data set with 76 variables (controllable, noncontrollable, and response, except of the boiler efficiency) as predictors.

SUMMARY PASSAGE 8

D. Comparison Of Coupling Effects For Two Clustering Strategies

The model S for predicting boiler efficiency, megawatt load, and unit heat rate was built separately. S was then used to predict the boiler efficiency, megawatt load, or unit heat rate for the controlled test data set. The "Data set 2" is the training data set from extracting centroids.

SUMMARY PASSAGE 9

E. Implementation Of The Clustering-Based Combustion Efficiency Controller

The clustering algorithm learns and generates new knowledge used to update the control signature database. Based on the real-time boiler status, the optimization algorithm searches the control signature database for an optimal centroid controlling the process. Thus, the boiler performance is improved.

SUMMARY PASSAGE 10

Vii. Conclusion

In this paper, a data-mining approach was applied to optimize boiler efficiency while considering coupling effects. Computational experiments showed that using 76 variables to construct centroids to control boiler efficiency reduced variance in the megawatt load and the unit heat rate due to the inclusion of the response variables in the centroids.